Reproducible Criterion-Level Clinical Trial Screening with an Open-Weight Transformer

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https://github.com/taimoorahmadx/genai-a2

***Abstract*—This paper describes an educational prototype for classifying individual clinical-trial criteria against patient descriptions and returning sentence-level evidence. The implementation uses the required TrialGPT Criterion Annotations data, a deterministic patient-grouped split, an open-weight autoregressive model interface, parameter-efficient supervised adaptation, and explicit implementations of cached decoding, tiled online-softmax attention, and greedy speculative decoding. Colab Tesla T4 runs completed supervised adaptation and validation, and synthetic attention and reduced decoder smoke benchmarks. The held-out test was evaluated once after settings were frozen. Decoder timings use one repeat and are exploratory; this paper reports them with that limitation.**

***Index Terms*—clinical trial matching, language models, evidence retrieval, KV cache, online softmax, speculative decoding**

I. Introduction

Trial eligibility criteria are expressed as natural-language constraints, while patient records contain a mixture of relevant findings and missing information. TrialGPT studies criterion-level matching and trial ranking with language models [1]. This project implements an independent, locally executable educational system and focuses on traceable criterion labels, sentence IDs, and inference mechanics. It is not intended for clinical decisions.

II. Related Work

TrialGPT motivates separating criterion matching from downstream trial ranking [1]. Low-Rank Adaptation (LoRA) reduces the number of trainable parameters through low-rank updates [6]. Transformer inference can reuse prior key/value states during autoregressive decoding. FlashAttention demonstrates that tiling with online softmax can compute exact attention while reducing memory traffic [7]. Speculative decoding drafts several tokens and verifies them with a target model while preserving greedy output under the corresponding greedy verification rule [8]. The selected Qwen2.5 model cards document the target and draft models' architecture, parameter counts, tokenizer and context length [3], [4].

III. Materials and Methods

A. Dataset and split

The benchmark is `ncbi/TrialGPT-Criterion-Annotations`, containing 1,015 criterion annotations and expert labels/evidence IDs [2]. The preprocessing allowlists patient note, criterion type/text, expert category, and expert sentence IDs, excluding generated GPT-4 fields. Unique patient identifiers are sorted, permuted with NumPy `default_rng(42)`, then assigned in groups of 37/8/8 to train, validation, and test. The program verifies coverage and non-overlap. Test examples are not materialized during preparation.

B. Model and adaptation

The default target is Qwen2.5-1.5B-Instruct: a 1.54B decoder-only model with RoPE and grouped-query attention, advertised 32,768-token context and Apache-2.0 license [3]. Qwen2.5-0.5B-Instruct (0.49B) is the draft candidate; the Qwen2.5 family uses a unified tokenizer, and code checks actual vocabularies before speculative decoding [4]. Phi-3.5-mini-instruct (3.8B, MIT, 128K context) is a larger alternative with higher expected weight and optimizer memory requirements [5]. The repository environment is CPU-only (PyTorch 2.14.1), while the reported fine-tuning run was performed on a Colab Tesla T4. The training script applies LoRA to attention projections with rank 16, alpha 32 and dropout 0.05. It trains for three epochs with learning rate 2×10^{-4}, batch size one, eight gradient-accumulation steps and 4096 token maximum. Loss is masked on prompt tokens and applied to the serialized answer; checkpoint selection uses validation loss. Colab required FP16 on the T4 and a Transformers 5 compatibility edit; these edits were made only in the runtime and must be reconciled for exact repository reproduction.

The model receives the patient note, criterion type and criterion text. It must return one of six exact expert categories: included, not included, not enough information, excluded, not excluded, or not applicable, plus sentence IDs and a concise explanation. Only IDs present in the parsed source note are retained. Truncation preserves the suffix of the prompt under the token budget; truncation can discard earlier note content and therefore is a limitation to quantify.

C. Evaluation and inference algorithms

Classification uses accuracy, macro-F1, per-category precision/recall/F1 and confusion matrices overall and by inclusion/exclusion. Evidence is scored by sentence-ID micro and macro precision/recall/F1; empty/empty per-record evidence sets receive one for precision and recall, while one-sided empty sets receive zero. A patient-level outcome, if shown, is an explicitly stated rule aggregation and is not a dataset label.

The cached decoder prefills the prompt once, then forwards the newest token with the returned cache, an absolute position ID, and an attention mask covering cached and current positions. The uncached reference recomputes the full sequence each step. Tiled attention keeps running row maxima m, softmax normalizers l, and weighted-value accumulators o while processing key blocks; outputs are checked against a direct scaled dot-product implementation. Greedy speculative decoding drafts a token block, verifies candidates with target logits, accepts the matching prefix, appends the target correction at first disagreement, and synchronizes cache state by re-prefilling emitted context. This conservative re-prefill favors clear state correctness and can reduce speed.

D. Application

The Streamlit application accepts numbered patient text and free-text inclusion/exclusion criteria, displays criterion category, explanation and evidence text, exposes a transparent aggregation rule and downloads a JSON report. It includes an educational-use notice and requires the local model to load successfully.

Figure 1 summarizes the supervised data path and application inference path. Test patients were kept untouched until the one-time final evaluation. The app parses model output and filters evidence IDs against the source note before display and export.

IV. Experimental Setup and Results

Training, validation evaluation, and synthetic attention and decoder benchmarks have been run in Colab; the final test was evaluated once on Kaggle after settings were frozen. The same 206 validation annotations were evaluated with the base model and adapter. Table I reports validation results. The adapter increased validation accuracy from 0.262 to 0.723 and macro-F1 from 0.111 to 0.356. Its predictions remain strongly biased: it predicted *not excluded* for all 140 validation exclusion examples, failing to recover the two gold *excluded* and eleven *not applicable* cases. Validation exclusion macro-F1 was 0.140. On the held-out test set, adapter accuracy was 0.689, macro-F1 0.344, and evidence micro-F1 0.442 (Table III). Macro-F1 is the unweighted mean over the six predefined labels, assigning zero to undefined class scores. Table II gives per-class scores and the confusion matrix. For all 131 test exclusion examples, the model predicted *not excluded*; this correctly classified 91, but misclassified 28 *not enough information* and 12 *not applicable* examples. On test inclusion criteria, accuracy was 0.680 and macro-F1 0.235. Inclusion/exclusion macro-F1 averages over the same six predefined labels, assigning zero to undefined class scores, including labels absent from a subset. The test set contained no gold *excluded* examples, so test performance on that class cannot be assessed. No settings were changed using test results. Row-level evaluation artifacts remain in Kaggle and are not committed because they contain patient notes. On six small synthetic causal/noncausal attention cases, tiled outputs matched the reference with maximum absolute error 5.36×10^{-7}, but the tiled Python implementation was 2.58–51.98 times slower; these measurements do not establish an advantage on model workloads. For three short synthetic decoding cases, uncached, KV-cached, and speculative generation produced identical token IDs. KV cache was 1.85–4.06 times faster than uncached generation. Speculative decoding accepted all proposed tokens in these short repetitive outputs, but was 1.56–2.08 times slower than KV caching. These decoder timings used a single measured repeat, so they are exploratory; longer repeated measurements remain pending; target and draft tokenizer vocabularies and special token IDs were verified to match.

The three-epoch training run completed 228 optimizer steps; the reported final training loss was 0.1347 and validation loss was 0.04293. The per-step training log was not retained, so a learning-curve figure cannot be produced from the available records.

TABLE I
BASELINE AND LORA ADAPTER VALIDATION RESULTS.

Measure	Base	LoRA adapter
Accuracy	0.262	0.723
Macro-F1	0.111	0.356
Evidence micro-F1	0.000	0.429
Inclusion accuracy	0.424	0.727
Inclusion macro-F1	0.116	0.250
Exclusion accuracy	0.186	0.721
Exclusion macro-F1	0.080	0.140

V. Limitations and Failure Analysis

The task's 53-patient sample is small; both validation and test show strong class bias. Criterion labels are not independent across patients. In a targeted review of eight criterion-level validation errors drawn from only three patient notes, all eight gold 'not enough information' exclusion labels were predicted as 'not excluded'. On test, every exclusion criterion was predicted as 'not excluded', including 28 'not enough information' and 12 'not applicable' gold cases; there were no gold 'excluded' test examples. Several reviewed validation predictions cited unrelated sentences; another missed the annotated uterine-imaging evidence and cited symptoms instead. These small samples illustrate failure modes but cannot estimate their prevalence. Free-form generation can produce malformed or unsupported outputs despite JSON prompting and ID validation. Evidence annotations can be empty, and absent evidence must not be interpreted as proof of eligibility or ineligibility. The repeated decoder benchmark, report PDF, screenshots, model artifact publication, and demo recording remain pending; target and draft tokenizer vocabularies and special token IDs have been verified to match.

VI. Conclusion

The repository provides a reproducible implementation scaffold covering data integrity, supervised adaptation, evidence-aware evaluation, model inference algorithms, and a browser

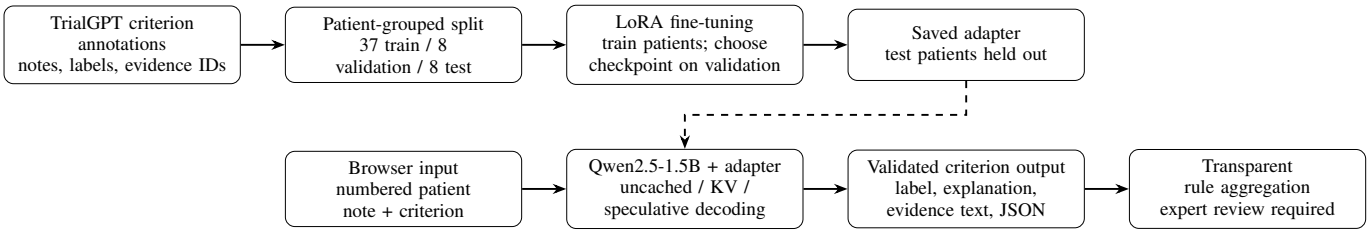


Fig. 1. System overview. Patient-grouped training produces a LoRA adapter; the application combines it with the base model and validates cited sentence IDs before showing criterion-level outputs. Tiled online-softmax attention is evaluated separately against reference attention.

TABLE II
HELD-OUT TEST PER-CLASS SCORES AND CONFUSION MATRIX (ROWS ARE GOLD LABELS; COLUMNS ARE PREDICTED LABELS).

	Per-class scores				Confusion matrix					
Class	P	R	F1	n	Inc.	Not inc.	NEI	Exc.	Not exc.	N/A
Included	1.000	0.481	0.650	27	13	0	14	0	0	0
Not included	0.000	0.000	0.000	7	0	0	7	0	0	0
Not enough information	0.613	0.576	0.594	66	0	0	38	0	28	0
Excluded	0.000	0.000	0.000	0	0	0	0	0	0	0
Not excluded	0.695	1.000	0.820	91	0	0	0	0	91	0
Not applicable	0.000	0.000	0.000	15	0	0	3	0	12	0

TABLE III
ONE-TIME HELD-OUT TEST RESULTS FOR THE FINE-TUNED ADAPTER.

Measure	Overall	Inclusion / exclusion
Accuracy	0.689	0.680 / 0.695
Macro-F1	0.344	0.235 / 0.137
Evidence micro-F1	0.442	–
Annotations	206	75 / 131

TABLE IV
PRELIMINARY DECODER SMOKE BENCHMARK ON COLAB T4; SECONDS, ONE REPEAT.

Input/output tokens	Uncached	KV cache	Speculative
32/4	0.368	0.199	0.310
64/4	0.557	0.205	0.324
128/8	1.717	0.423	0.880

application. Validation improved over the base model on aggregate metrics, but the one-time test results confirm substantial class bias, especially on exclusion criteria. The inference smoke tests show cached decoding gains and no speculative gain on the tested short outputs. A repeated decoder benchmark remains needed. The system remains an educational prototype and cannot determine enrollment eligibility.

AI USE ACKNOWLEDGMENT

AI assistance supported repository scaffolding and implementation. The student must review and understand the code, perform the experiments, verify sources, and disclose AI use in the submitted work.

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